

Day 3 — Transformers for Vision

From attention to ViT, and how it compares to CNNs

The Transformer block

The original Transformer
(Vaswani et al., 2017):

- **Multi-head self-attention**
— every token attends to every other token
- **Feed-forward MLP** applied per token
- **Residual + LayerNorm** around each sub-block
- Stacks of these blocks, no recurrence, no convolution

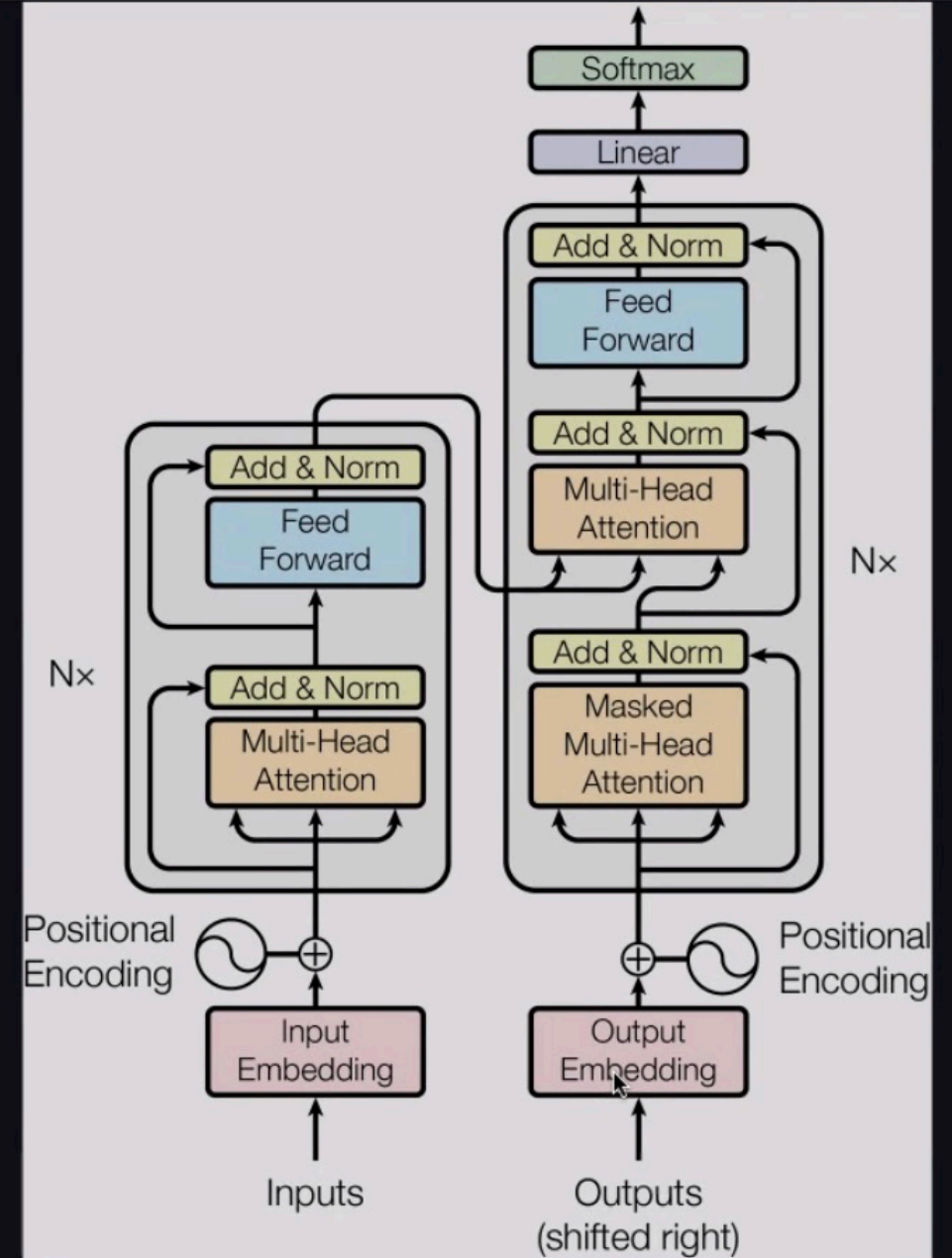


Figure 1: The Transformer - model architecture.

Self-attention in one line

For each token, compute a weighted sum of *all* tokens:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

- Q, K, V are linear projections of the input
- $QK^\top$ is the similarity matrix (who looks at whom)
- The softmax row gives attention weights summing to 1
- Multi-head = run this in parallel with different projections, then concatenate

From text tokens to image patches

Transformers eat sequences of tokens. An image is a 2D grid of pixels. **How do we bridge?**

→ Cut the image into fixed-size patches (e.g. 16×16), flatten each patch into a vector, and treat each patch like a "word".

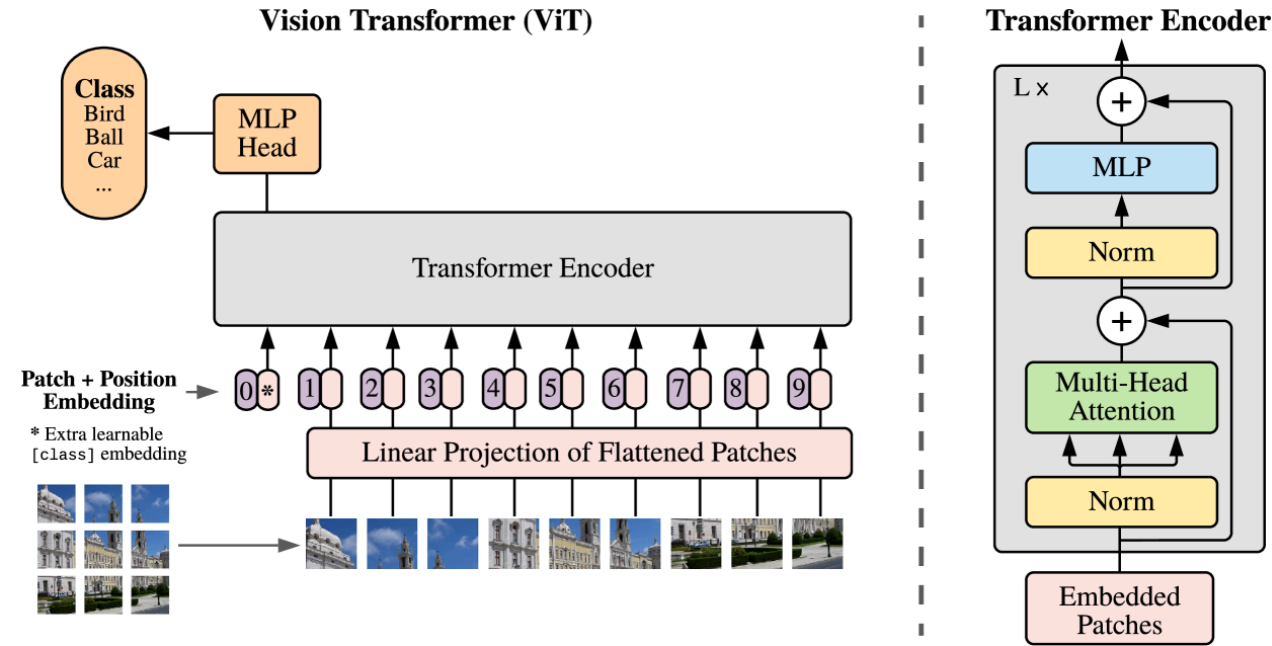
A 224×224 image with 16×16 patches → $14 \times 14 = 196$ **tokens**.

That's it — the rest of the architecture is essentially the same Transformer encoder.

Vision Transformer (ViT)

Dosovitskiy et al., 2020 — *"An image is worth 16×16 words"*

1. Split image into patches
2. **Linear projection** of each patch
→ patch embedding
3. Prepend a learned [CLS] token
4. Add **positional embeddings** (the model has no spatial prior!)
5. Pass through N Transformer encoder blocks



What's different from a CNN?

Aspect	CNN	ViT
Inductive bias	Locality, translation equiv.	Almost none
Receptive field	Grows with depth	Global from layer 1
Parameters	Shared filters	Dense attention + MLPs
Data hunger	Works on small datasets	Needs lots of data (or pretraining)
Compute	FLOPs $\propto$ spatial size	FLOPs $\propto$ (num patches) ² for attention

When does each win?

CNNs win when:

- Data is limited ($< \sim 1\text{M}$ images)
- Strong locality matters (textures, fine-grained edges)
- You need cheap inference at high resolution

ViTs win when:

- You have a lot of data — or a good pretrained model
- The task benefits from long-range dependencies
- You're fine-tuning from a foundation model (DINOv2, CLIP, SAM ...)

Hybrid models (Swin, ConvNeXt) borrow ideas from both.

What you'll build today

Notebook `09_building_vit.ipynb` — a ViT from scratch:

1. **Patch embedding** with `nn.Conv2d(kernel_size=patch, stride=patch)`
2. **Learnable [CLS] token + positional embedding**
3. A single **Transformer encoder block** (multi-head attention + MLP + residuals)
4. Stack N blocks, classification head
5. Train on CIFAR-10 — confirm it learns, even if accuracy lags a CNN at this scale

Then `10_cnn_vs_vit.ipynb` — side-by-side comparison.

Let's build a ViT.

Open `notebooks/09_building_vit.ipynb`